

DESIGN PROBLEM

As bungee advocates we stand that bungees are the best power source. We believe that the bungee is the answer to all transportation problems. Hence we produced a bungee powered vehicle that drives across both a flat and bumpy track, and travels up a steep ramp whilst carrying a payload.

Seen in Fig. 1 below are the events that our vehicle is tasked to complete all while fitting within the constraints outlined in the section below.

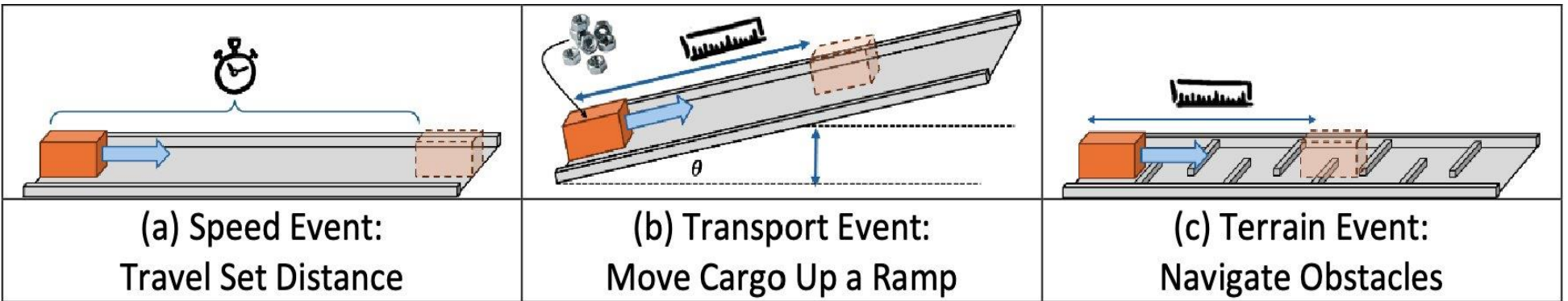


Figure 1: Problem Summary

PROJECT CONSTRAINTS

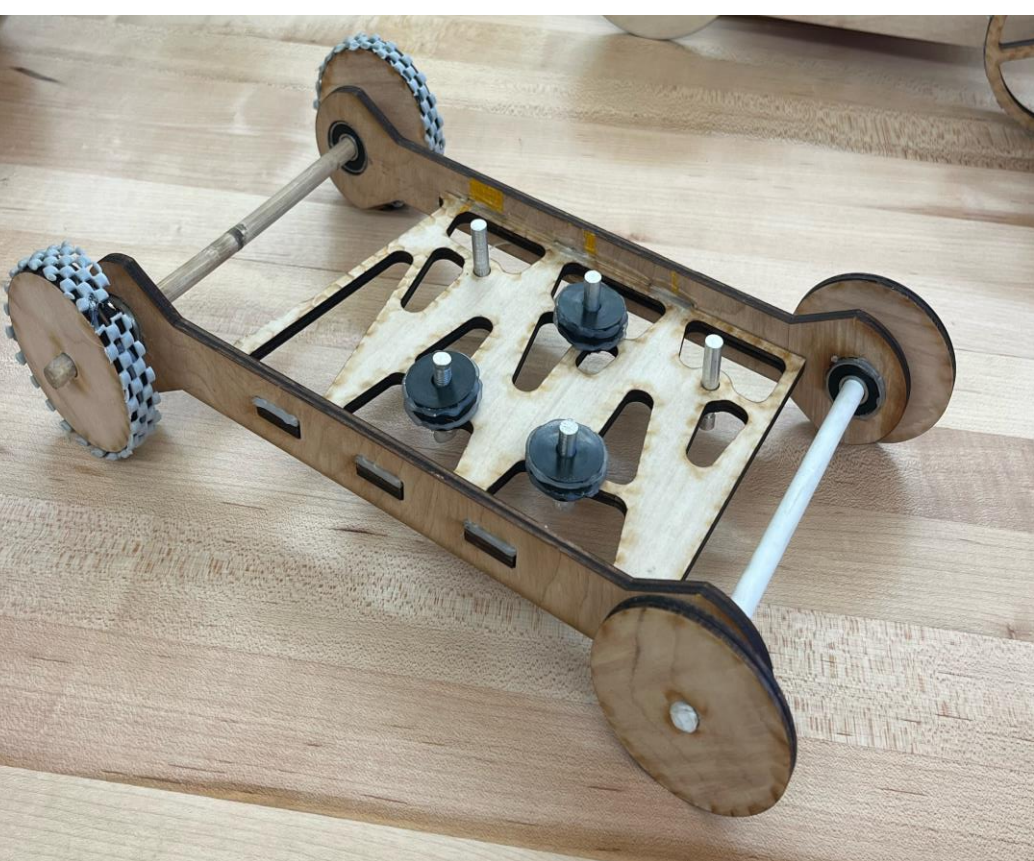
- 100% Mechanical
- Limited Material options + associated cost
- Self Contained
- Size Constraint
 - 12' x 11' x 12' (L x W x H)
- Safety
 - For both operator(s) and spectators

PROTOTYPES



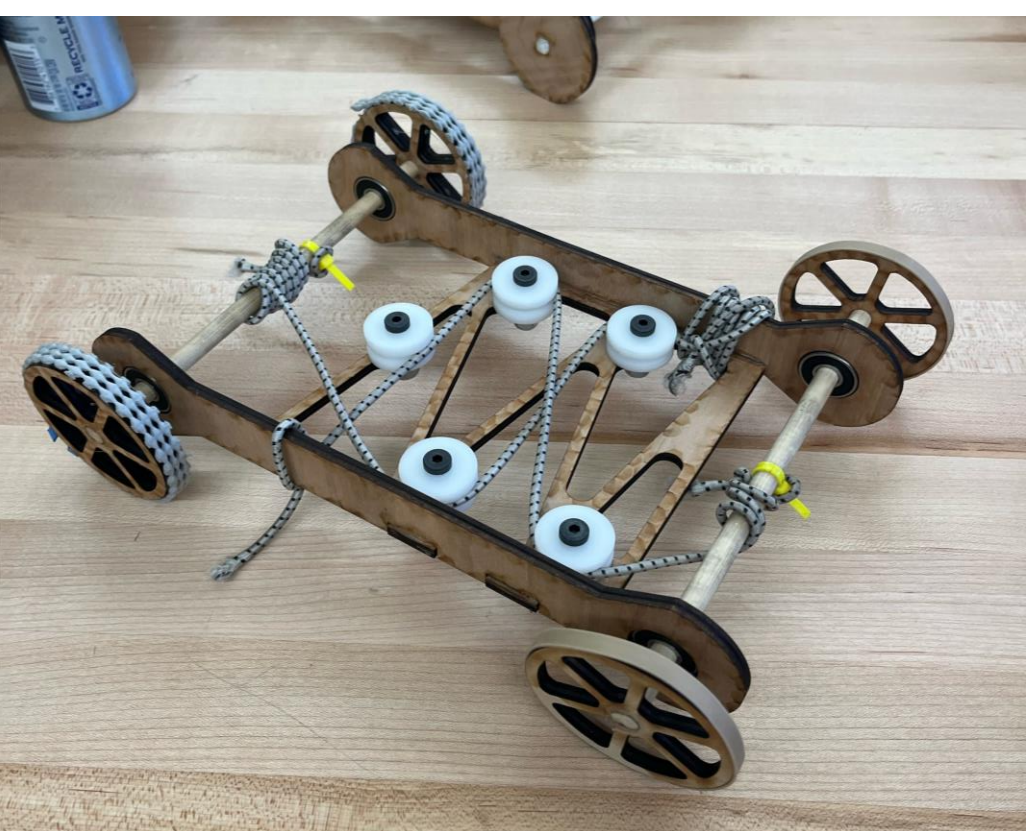
- Prototype v1:
- Rudimentary materials
 - Verify bungee cord as viable power source
 - Proof of Concept

Figure 2: Prototype v1



- Prototype v2:
- Lightweight Frame
 - Improved pulleys
 - Smaller Wheels
 - Tire Traction Mod

Figure 3: Prototype v2



- Prototype v3:
- 4WD System
 - Grooved pulleys
 - Lightweight Wheels

Figure 4: Prototype v3



Prototransmission:
Our unsuccessful attempt to create torque for 4" wheels

Ask why it was unsuccessful!

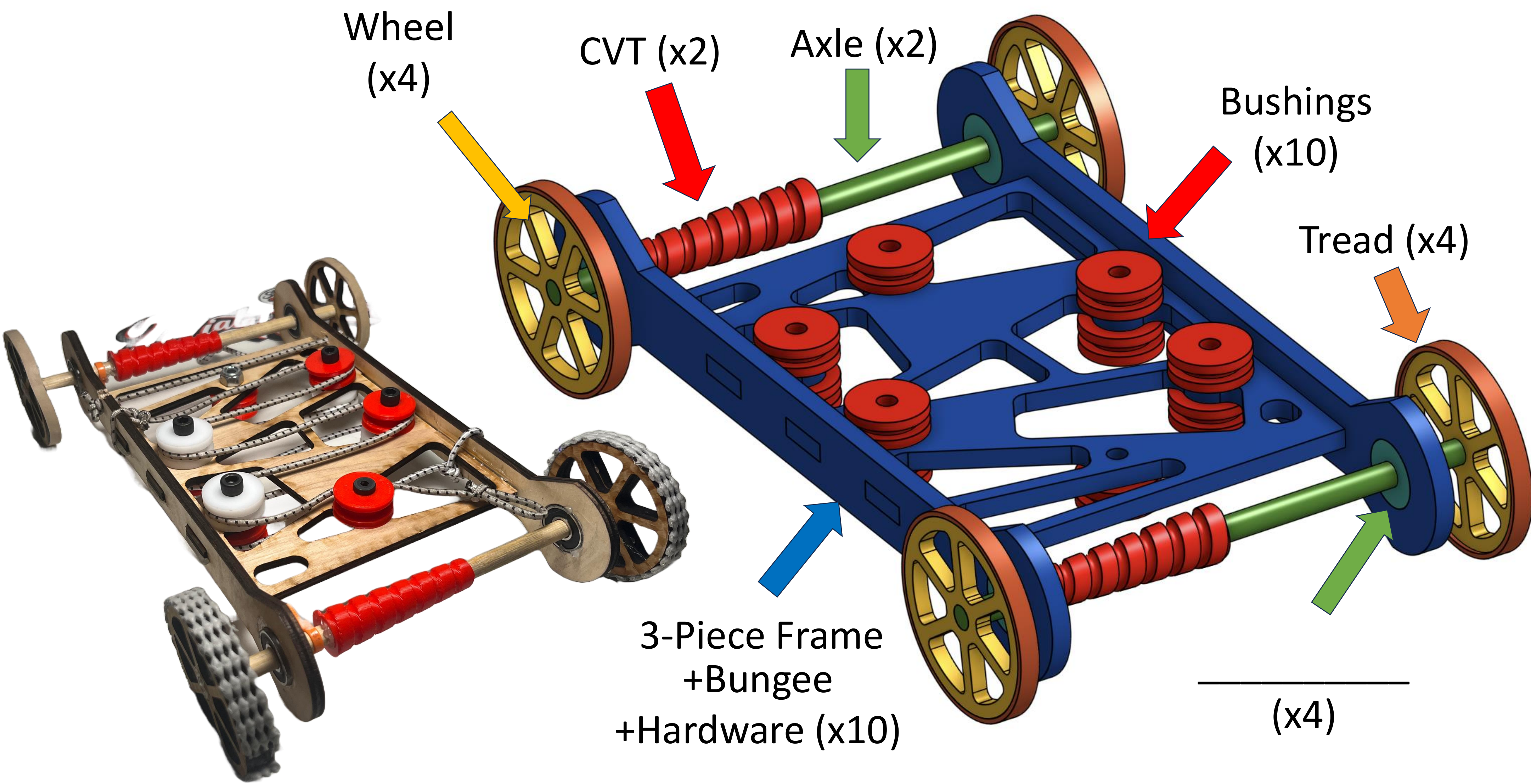
Figure 3.5: Prototransmission

The Bungee Mobile

TEAM 109

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DESIGN SOLUTION



FRAME

- Frame design:
- 2 side rails
 - 1 central main frame
 - Cutouts to integrate guided pulley system
 - Weight reduction hollowing
- Frame Goal:
- Reduce Friction
 - Maximize Power Transfer
 - Handle the stresses of ALL the design aspects (axle + bungee + pulleys)

Refer to prototypes and design solution for visuals

Ask us how we did the weight reduction!

WHEELS

- Wheel size:
Reducing the wheel diameter from 4" to 2.5", we make the wheel easier to turn at the cost of having to turn it more to go 8'. For us, it was worth it.
- Traction
 - A soft mesh material wrapped around the wheel improves grip, maximizing our bungees energy



Figure 5: Traction padding

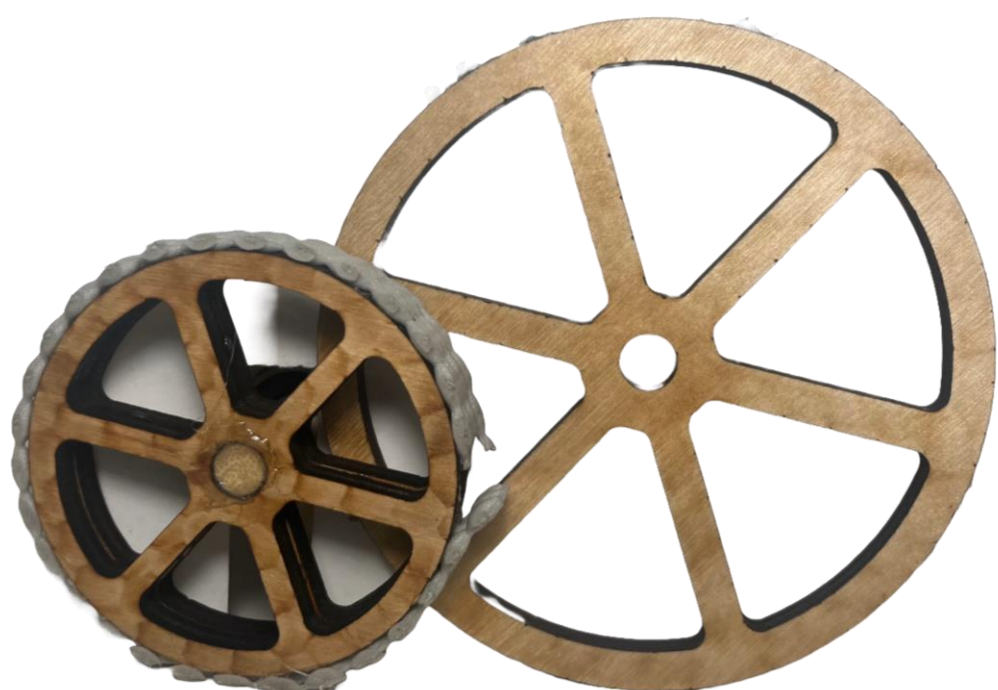


Figure 6: Wheel Comparison

Drive Train

- CVT optimizes torque output
- 4WD system utilizes 1 power source per axle
- Bungee cord systems run on top and underneath the car, connecting and delivering power to both axles.

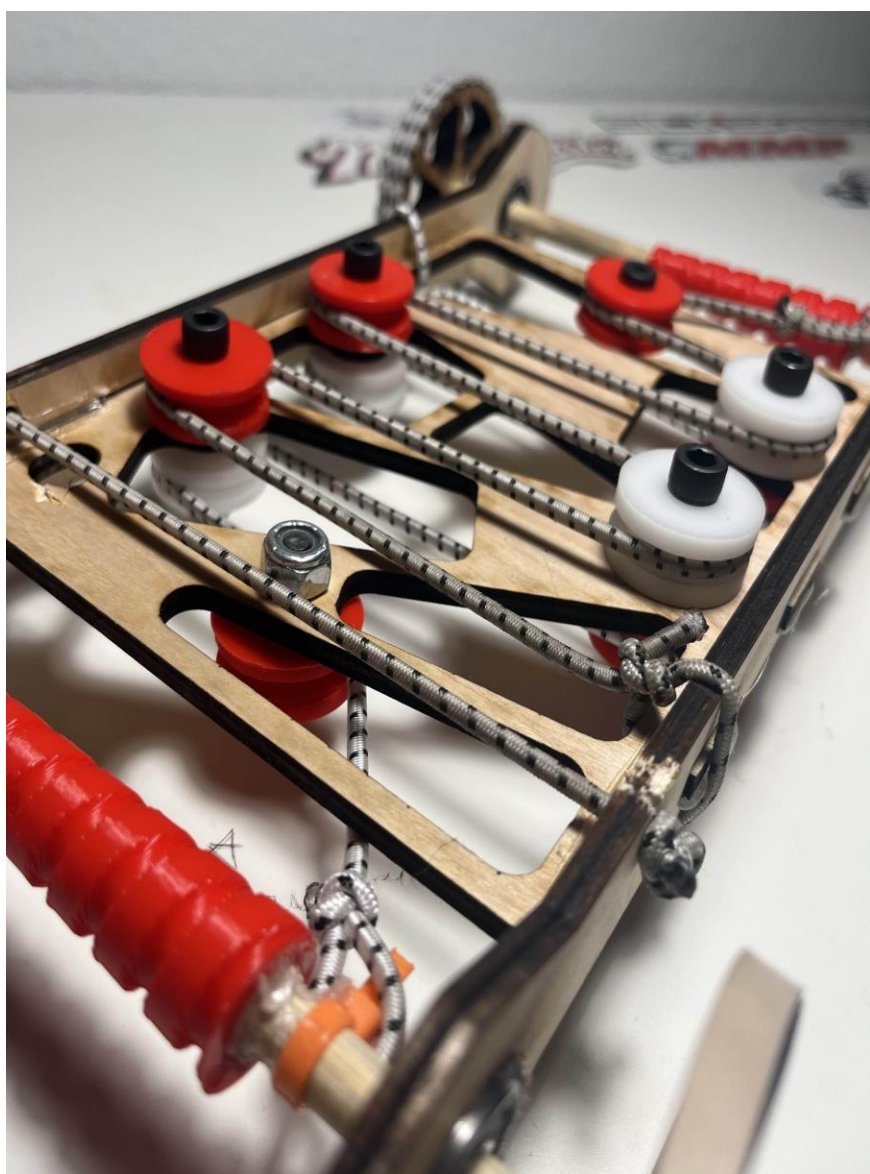


Figure 8: 4WD System

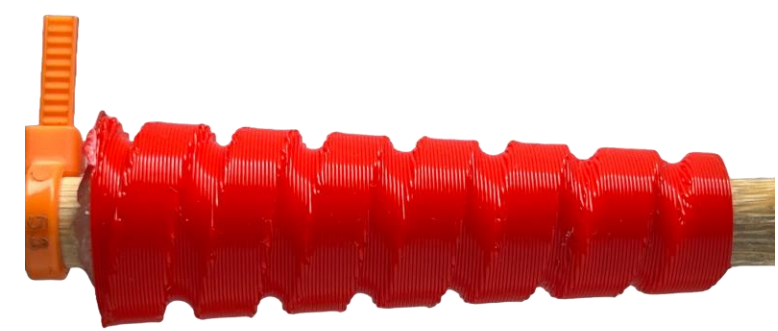


Figure 9: CVT Cone

PULLEYS

- Wound pulleys allowing for maximum bungee length
- Freely spinning pulleys **ideally** create even tension throughout the whole cord when loading.
- (we are not ideal, but its good enough)
- The last pulley is centered within the car to give a straight on release to the axle

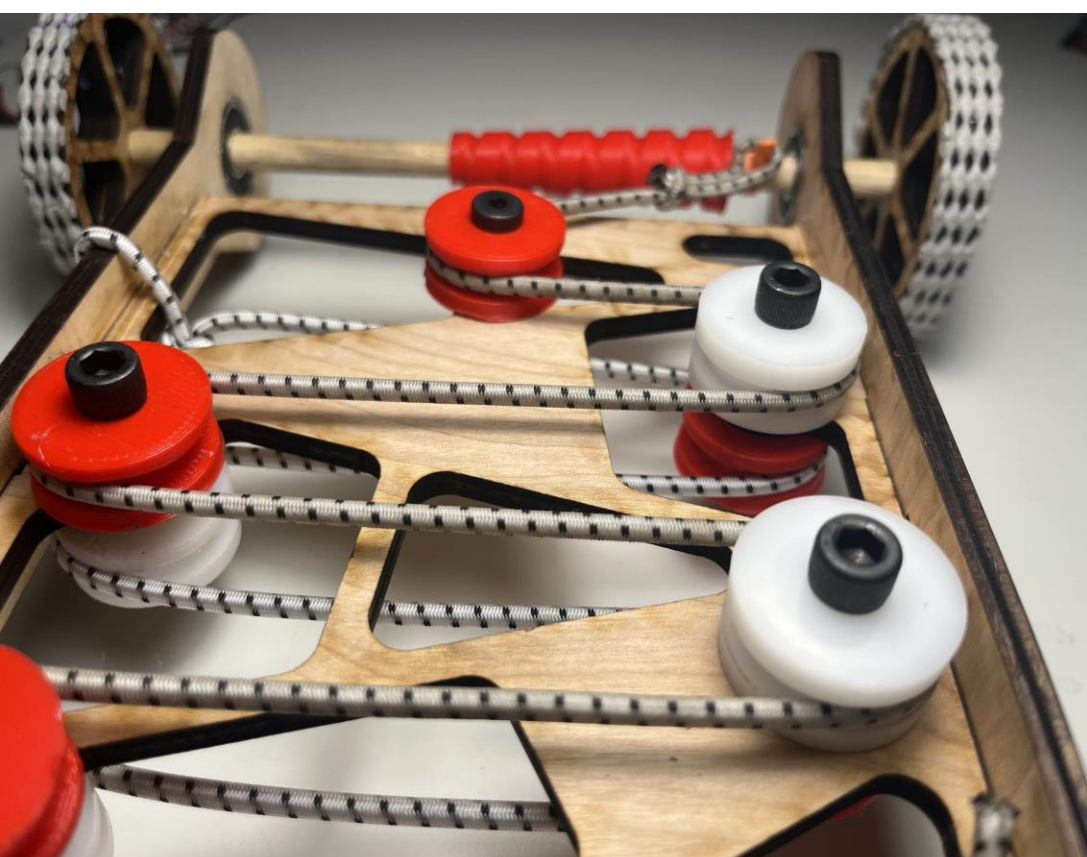


Figure x: Pulley system 7



Economics & Assembly Details

1. Wheels Material: Wood Cost: \$0.83	2. CVT Material: PLA Cost: \$0.42	3. Axle Material: Wood Cost: \$0.75
4. Frame Material: Wood Cost: \$1.66	5. Bushings Material: PLA Cost: \$1.20	6. Bungee Material: Bungee** Cost: \$1.10
6. Hardware Material: ¼ -20* Cost: \$0.60	8. Tread Material: _____ Cost: \$ _____	9. _____ Material: _____ Cost: \$ _____

*4 @ 1½" length & 2 @ 1"
**5.5 feet of bungee

Grand Total: \$ _____

Ask us how we cut some costs!

BACKGROUND / MARKET RESEARCH

Originally, we were going to go with a spring design where the axle is powered by the compression of a spring.

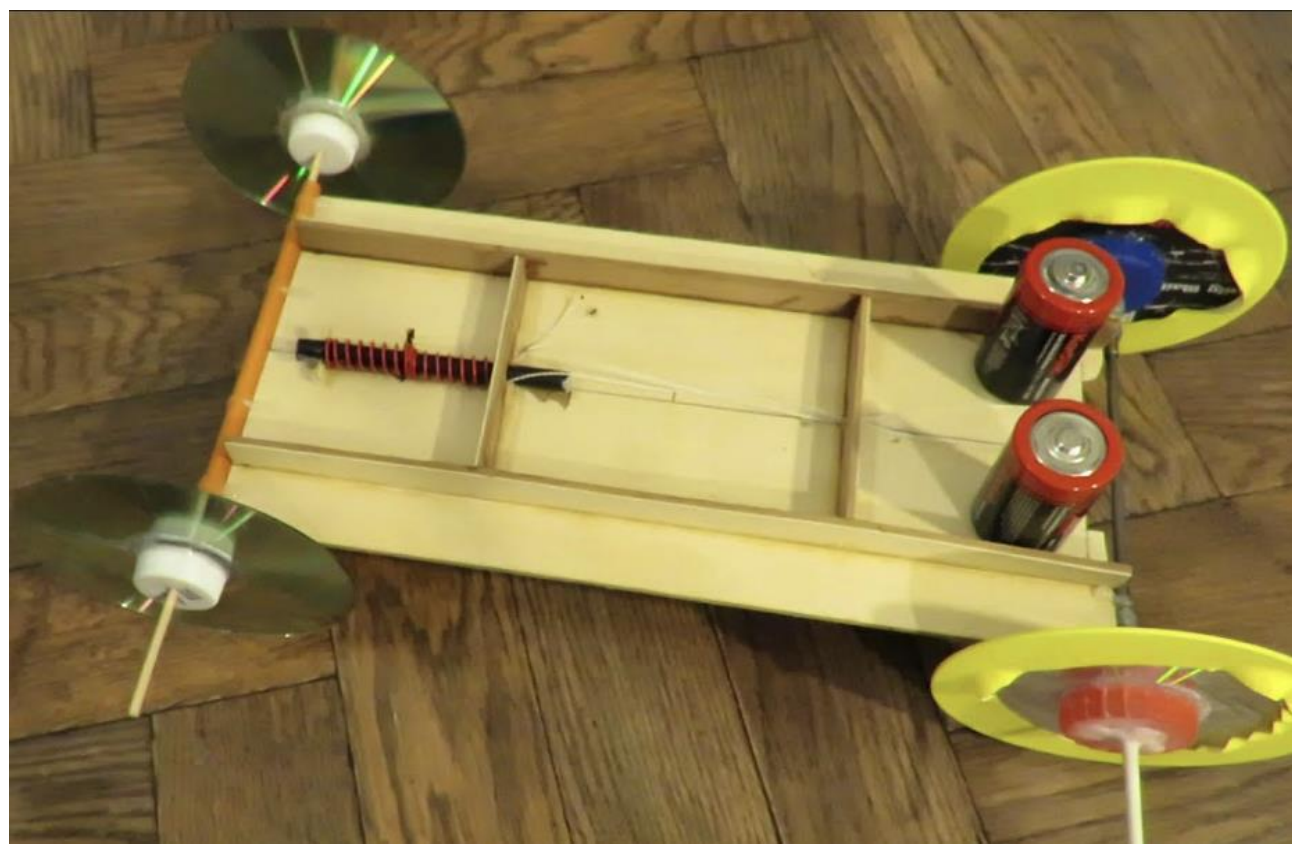


Figure 10:
Compression Spring
Design Inspiration

"Spring powered car". YouTube.
www.youtube.com/watch?v=bfrjsODECr&ab_channel=GrandadIsAnOldMan

We struggled early on with friction issues and quickly pivoted believing this solution to not a big competitor in the market.

On the other hand, solutions like flywheels and torsion springs are present and competitive given simplicity and high energy conversion.

Ask Us why we went with our design!

LESSONS & FUTURE WORK

Looking back on our project and design, we took a lot away from our failures. And by failing fast we were able to ultimately succeed. INNOVATE! Learn and move forward.

Looking forward, the next steps would be to scale the forces that our bungee can output with the upper limits that our frame, pulley, and axle can withstand in order to get the most out of our design. Or continue to find ways to minimize costs through cheaper materials.